



HR Analytics Automation Platform

Version 1.0

Data Analytics Portfolio Project

Python | Excel | Power BI | xlwings

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Professional Profiles

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1. Executive Summary

The HR Analytics Automation Platform is an end-to-end workforce analytics solution developed using Python, Excel, Power BI, and automation technologies. The project provides actionable insights into employee attrition, workforce demographics, compensation trends, and employee satisfaction through interactive dashboards and advanced analytics.

2. Project Overview

The HR Analytics Automation Platform was developed to analyze employee attrition, workforce trends, compensation patterns, and employee satisfaction using Python, Excel, xlwings, and Power BI.

3. Business Objective

The primary objective of this project is to analyze workforce trends, employee attrition, compensation patterns, and job satisfaction metrics to support data-driven HR decision-making through automated reporting and interactive dashboards.

4. Dataset Overview

Metric	Value
Total Employees	1470
Departments	3
Key Variables	35+
Target Variable	Attrition
Data Source	IBM HR Analytics Dataset

5. Tools & Technologies

- Python
- Pandas
- Excel
- xlwings
- Power BI
- DAX
- Seaborn
- Matplotlib

6. Data Preparation

- Column standardization
- Creation of Attrition Flag
- Creation of Income Band
- Creation of Tenure Group
- Data validation and cleaning

7. KPI Analysis

Total Employees

Measures the total workforce population included in the analysis.

Attrition Rate

Represents the percentage of employees who left the organization.

Average Salary

Provides insights into overall workforce compensation.

Average Tenure

Measures employee retention and workforce stability.

8. Dashboard Development

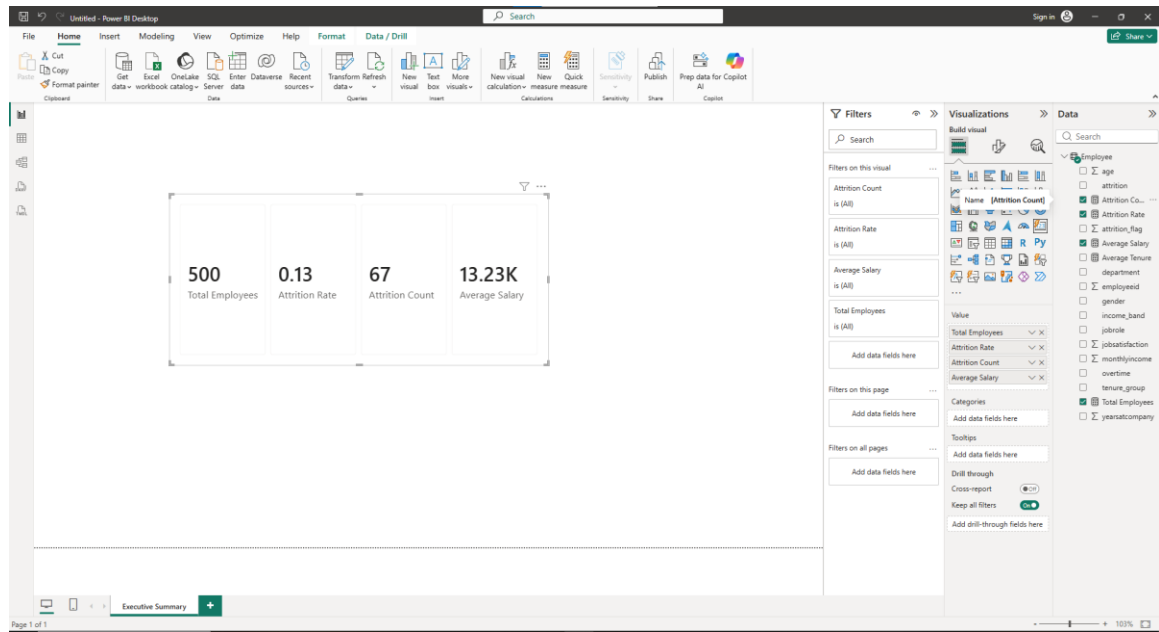
Interactive dashboards were developed using Excel and Power BI for workforce analytics.

Dashboard Features

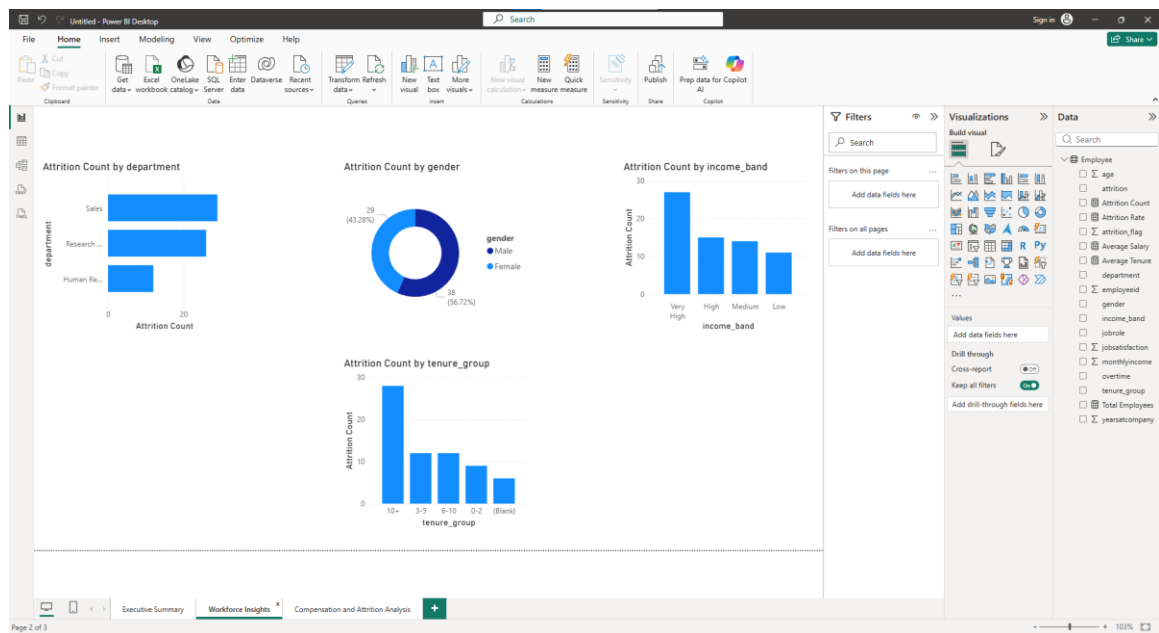
- KPI Monitoring
- Workforce Segmentation
- Attrition Analysis
- Compensation Analysis
- Interactive Slicers
- Python Integration

HR Analytics Automation Platform

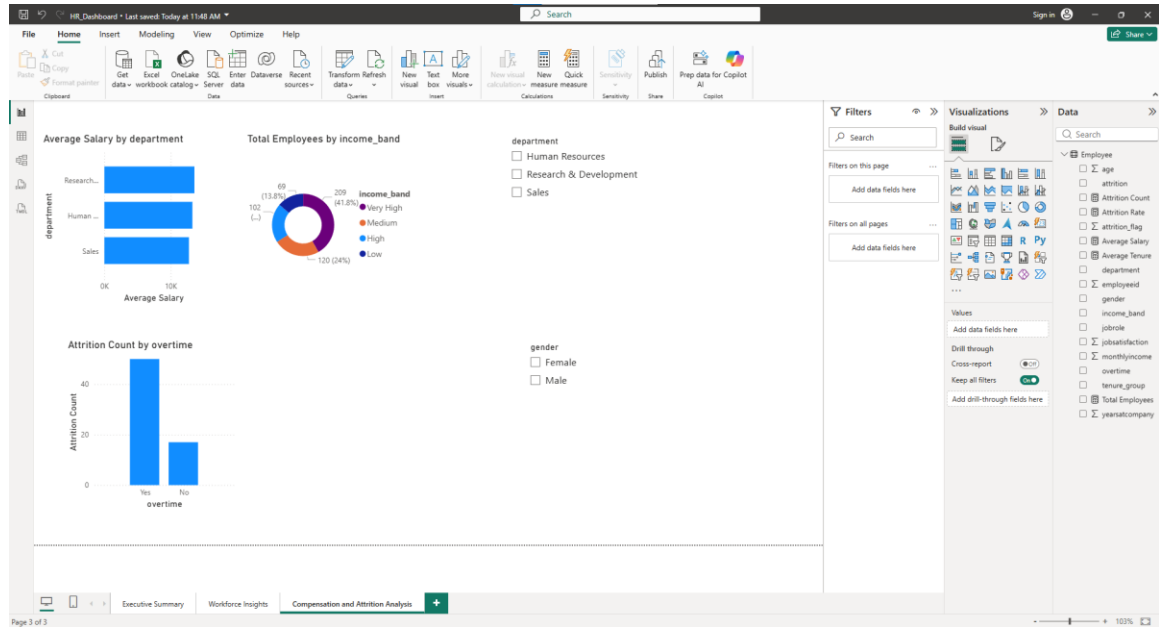
Executive Summary Screenshot



Workforce Insights Screenshot



Compensation & Attrition Screenshot



9. Project Architecture

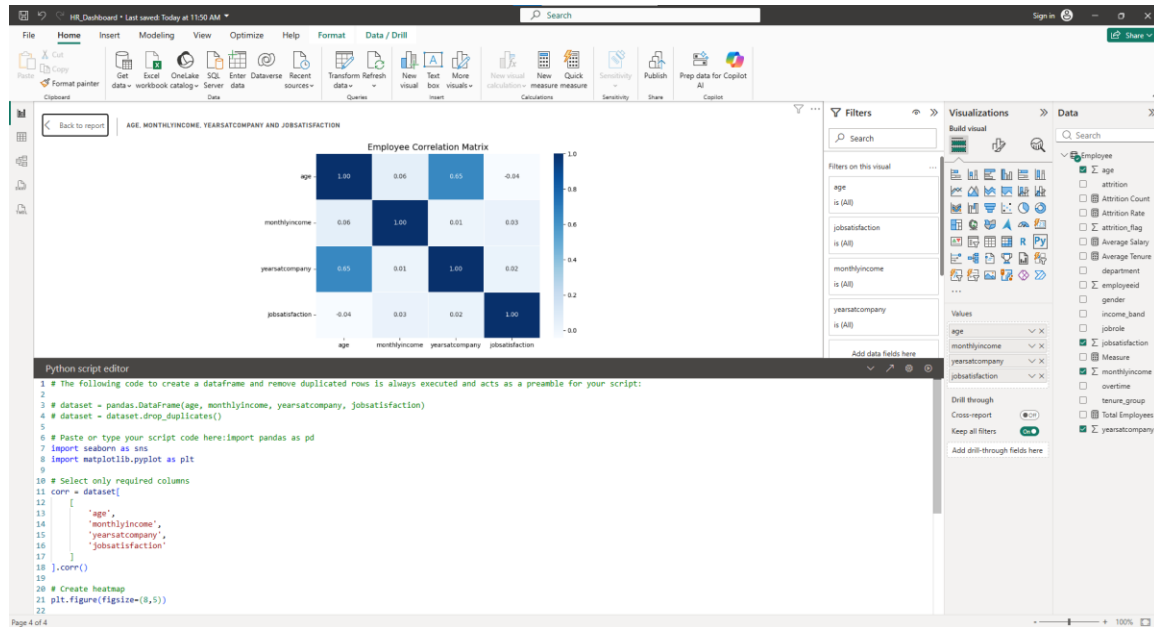
The HR Analytics Automation Platform follows an end-to-end analytics workflow integrating data preparation, automation, dashboard, and advanced analytics.



10. Advanced Analytics

A Python-based correlation heatmap was developed using Seaborn and Matplotlib to analyze relationships between employee age, salary, tenure, and job satisfaction.

Employee Correlation Matrix



A correlation heatmap was developed using Pandas, Seaborn, and Matplotlib to analyze relationships between employee age, compensation, tenure, and job satisfaction.

11. Key Insights

- Sales department exhibited the highest attrition rate.
- Employees working overtime demonstrated higher attrition tendencies.
- Salary increases positively correlated with employee tenure.
- Job satisfaction displayed weaker relationships with compensation factors.

12. Recommendations

- Reduce overtime burden.
- Strengthen retention programs.
- Monitor workforce satisfaction.
- Promote data-driven HR decisions.

13. Future Enhancements

- Predictive Attrition Modeling using Machine Learning.
- Automated Email Reporting for HR stakeholders
- Real-Time Dashboard Refresh and Monitoring.
- Cloud Deployment using Microsoft Azure Services.
- Integration with HR Management Systems for live workforce analytics.

14. Conclusion

The project successfully demonstrates the integration of Python, Excel, automation, and Power BI to support HR analytics and workforce decision-making.